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Solar Cell Efficiency

The efficiency is the most commonly used parameter to compare the performance of one solar cell to another. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. In addition to reflecting the performance of the solar cell itself, the efficiency depends on the spectrum and intensity of the incident sunlight and the temperature of the solar cell. Therefore, conditions under which efficiency is measured must be carefully controlled in order to compare the performance of one device to another. Terrestrial solar cells are measured under AM1.5 conditions and at a

temperature of 25°C. Solar cells intended for space use are measured under AM0 conditions. Recent top efficiency solar cell results are given in the page [Solar Cell Efficiency Results](#) .



Testing silicon solar cells.

The efficiency of a solar cell is determined as the fraction of incident power which is converted to electricity and is defined as:

$$P_{\max} = V_{OC} I_{SC} FF$$

$$\eta = \frac{V_{OC} I_{SC} FF}{P_{in}}$$

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Equations

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Where:

V_{oc} is the open-circuit voltage;

I_{sc} is the short-circuit current;

FF is the fill factor and

η is the efficiency.

The input power for efficiency calculations is 1 kW/m^2 or 100 mW/cm^2 .

Thus the input power for a $100 \times 100 \text{ mm}^2$ cell is 10 W and for a $156 \times 156 \text{ mm}^2$ cell is 24.3 W

Efficiency Calculator

Input Parameters

Open Circuit Voltage, V_{OC} (V)

0.611

Short Circuit Current, I_{SC} (A)

3.5

Fill Factor, FF

0.7

Input Power, P_{in} (W)

10

Results

Maximum Power, $P_{max} = 1.1762 \text{ W}$

Efficiency, $\eta = 14.9695 \%$

